Homework 3
Biostatistics 755
Due March 17th, 2026

A study was conducted to investigate two treatments for patients suffering from multiple sclerosis. 150 patients with the disease were recruited into the study, and 75 were randomized to receive azathioprine (AZ) alone (group 1), and 75 to receive azathioprine plus methylprednisolone (AZ+MP, group 2). For each participant, a measure of autoimmunity, azathioprine AFCR, was planned at clinic visits at baseline (time 0, at initiation of treatment) and at 3, 6, 9, 12, 15, and 18 months thereafter. Multiple sclerosis (MS) affects the immune system. Low values of AFCR (approaching 0) are evidence that immunity is improving, which is hopefully associated with a better prognosis for sufferers of MS. Also recorded for each subject was age at entry into the study and an indicator of whether or not the subject had had previous treatment with either of the study agents (0=no, 1=yes). The mean age of the men across both treatment groups was 50.45, with an SD of 6.69.

The primary scientific aims of the study are to investigate whether (i) both treatments (AZ or AZ+MP) lower AFCR over the 18 month period and (ii) whether treatment with AZ+MP results in a different immune system response than does AZ alone, and, if so, how it is different in terms of response over time. It was also suspected that a subject's age and prior history might be related to their baseline AFCR level and the rate of change in AFCR over the 18-month period.

The square root of AFCR is the response variable of interest (square roots were taken so that the AFCR observations better satisfy the assumption of normality).

The data are in the file `afcr`, which you can download from the course website. In the file, each record corresponds to a single observation, with columns:

```
column 1 = subject id
column 2 = time (months)
column 3 = square root AFCR
column 4 = group (1 = AZ alone, 2 = AZ + MP)
column 5 = prior treatment indicator
column 6 = age at baseline (yrs)
```

1. Discuss the issues of study design and how they would relate to your decision to use a linear mixed model or a pattern mixture model.

2. Regardless of your answer to (1), complete an analysis of this data using a linear mixed model. This analysis should yield the best possible answer to the study's scientific aims. This analysis should include:

   (a) Exploratory data analysis (e.g., figures) and discussion of findings.

   (b) How do we test the hypothesis of interest? Will you include any confounders or effect modifiers?

   (c) Model fit: compare different mean and random effect structures.

   (d) Model diagnostics checks.

(e) Identify the most influential subject? Are they influential due to their Y or X, and what are they mainly impacting?

(f) A paragraph that accurately summarized your conclusions with interpretations of coefficients of interest.

It is encouraged to give the code to replicate your findings.